

## Java Lab Programs

### Lab 1: Write a Java program to find factorial of numbers using command line arguments.

```
public class Lab1
{
    public static void main(String[] args)
    {
        int n,i,j,f;
        for(i=0;i<args.length;i++)
        {
            n=Integer.parseInt(args[i]);
            f=1;
            for(j=1;j<=n;j++)
            {
                f=f*j;
            }
            System.out.println("factorial of "+n+"="+f);
        }
    }
}
```

### Lab 2. Write a Java program to store and display student details using an array of objects.

```
import java.util.Scanner;
class Student
{
    String srn;
    String name;
    int marks;
    void getdata()
    {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter srn : ");
        srn=sc.next();
        System.out.print("Enter name : ");
        name=sc.next();
        System.out.print("Enter marks : ");
        marks=sc.nextInt();
    }
    void printdata()
    {
        System.out.print(srn+"\t"+name+"\t"+marks);
    }
}
```

```
class Lab2
{
    public static void main(String[] args)
    {
        int n,i;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter how many student information want to store : ");
        n=sc.nextInt();
        Student s[]=new Student[n];
        System.out.println("Enter student information : ");
        for(i=0;i<n;i++)
        {
            s[i]=new Student();
            s[i].getdata();
        }
        System.out.println("srn"+"\\t"+"name"+"\\t"+"marks");

        for(i=0;i<n;i++)
        {
            s[i].printdata();
        }
    }
}
```

### Lab 3A. Write a Java program to demonstrate method overloading.

```
class shape
{
    void area(double r)
    {
        double a=3.1425*r*r;
        System.out.println("Area of a Circle = "+a);
    }
    void area(int l,int b)
    {
        int a=l*b;
        System.out.println("Area of a Rectangle = "+a);
    }
    void area(float b,float h)
    {
        double a=0.5*b*h;
        System.out.println("Area of a triangle = "+a);
    }
}

public class Lab3A
{
    public static void main(String[] args)
    {
        shape s = new shape();
        s.area(2);
        s.area(3,4);
        s.area(3.5f,4.5f);
    }
}
```

### Lab 3B. Write a Java program to demonstrate constructor overloading.

```
class consover
{
    String srn;
    String name;
    int marks;
    consover()
    {
        srn= "unknown";
        name= "unknown";
        marks= 0;
    }
    consover(String s,String n,int m)
    {
        srn= s;
        name= n;
        marks= m;
    }
    void printdata()
    {
        System.out.println("srn= "+srn+"\t"+"name= "+name+"\t"+"marks= "+marks);
    }
}

public class Lab3B
{
    public static void main(String[] args)
    {
        consover o1=new consover();
        consover o2=new consover("123","xyz",40);
        o1.printdata();
        o2.printdata();
    }
}
```

#### Lab 4A.Aim: Write a Java program to demonstrate multilevel and hierarchical inheritance.

```
import java.util.*;
class A
{
    int m;
    int n;
    void getdata()
    {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter m,n :");
        m=sc.nextInt();
        n=sc.nextInt();
    }
    void printdata()
    {
        System.out.println("m= "+m+"\t"+"n= "+n);
    }
}
```

```
class B extends A
{
    int s;
    void sum()
    {
        s=m+n;
        System.out.println("Sum= "+s);
    }
}
```

```
class c extends B
{
    int p;
    void prod()
    {
        p=m*n;
        System.out.println("product= "+p);
    }
}
```

```
class D extends B
{
    int d;
    void div()
    {
        d=m/n;
        System.out.println("division= "+d);
    }
}
```

```

public class Lab4A
{
    public static void main(String[] args)
    {
        c x=new c();
        x.getdata();
        x.printdata();
        x.sum();
        x.prod();

        D y=new D();
        y.getdata();
        y.printdata();
        y.sum();
        y.div();
    }
}

```

**Lab 4B. Write a Java program to demonstrate method overriding.**

```

class AA
{
    void disp()
    {
        System.out.println("Welcome to DSA");
    }
}

class BB extends AA
{
    void disp()
    {
        System.out.println("Welcome to java");
    }
}

public class Lab4B
{
    public static void main(String[] args)
    {
        BB x=new BB();
        x.disp();
    }
}

```

## Lab 5. Write a Java program to demonstrate interfaces.

```
interface Triangle
{
    void area(double b,double h);
}
interface Circle{
    void area(double r);
}

class Areatc implements Triangle,Circle
{
    public void area(double b,double h)
    {
        double a=0.5*b*h;
        System.out.println("Area of a triangle="+a);
    }
    public void area(double r)
    {
        double a=3.142*r*r;
        System.out.println("Area of a circle="+a);
    }
}

public class Lab5
{
    public static void main(String[]args)
    {
        Areatc tc=new Areatc();
        tc.area(3,4);
        tc.area(2);
    }
}
```

## Lab 6A. Write a Java program to create threads using the Thread class.

```
class NaturalThread extends Thread
{
    public void run()
    {
        for (int i = 1; i <= 10; i++)
        {
            System.out.println("Natural Numbers: " +i);
            try
            {
                Thread.sleep(1000);
            }
            catch (InterruptedException e)
            {
                System.out.println("Exception");
            }
        }
    }
}
```

```
class EvenThread extends Thread
{
    public void run()
    {
        for (int i = 1; i <= 10; i++)
        {
            System.out.println("even Numbers: " +(i*2));
            try
            {
                Thread.sleep(1000);
            }
            catch (InterruptedException e)
            {
                System.out.println("Exception");
            }
        }
    }
}
```

```

class OddThread extends Thread
{
    public void run()
    {
        for (int i = 1; i <= 10; i++)
        {
            System.out.println("odd Numbers: " +(i*2-1));
            try
            {
                Thread.sleep(1000);
            }
            catch (InterruptedException e)
            {
                System.out.println("Exception");
            }
        }
    }
}

```

```

public class Lab6A
{
    public static void main(String[] args)
    {

        NaturalThread t1 = new NaturalThread();
        EvenThread t2 = new EvenThread();
        OddThread t3 = new OddThread();

        t1.start();
        t2.start();
        t3.start();
    }
}

```

## Lab 6B. Write a Java program to create threads using the Runnable interface.

```
class NaturalRunnable implements Runnable {
    public void run()
    {
        for (int i = 1; i <= 10; i++)
        {
            System.out.println("Natural Numbers: "+i);
            try
            {
                Thread.sleep(1000);
            }
            catch (InterruptedException e)
            {
                System.out.println("Exception");
            }
        }
    }
}

class EvenRunnable implements Runnable {
    public void run()
    {
        for (int i = 1; i <= 10; i++)
        {
            System.out.println("even Numbers: "+(i*2));
            try
            {
                Thread.sleep(1000);
            }
            catch (InterruptedException e)
            {
                System.out.println("Exception");
            }
        }
    }
}

class OddRunnable implements Runnable
{
    public void run()
    {
        for (int i = 1; i <= 10; i++)
        {
            System.out.println("Odd Numbers: "+(i*2-1));
            try
            {
                Thread.sleep(1000);
            }
            catch (InterruptedException e)
            {
                System.out.println("Exception");
            }
        }
    }
}
```

```
public class Lab6B
{
    public static void main(String[] args)
    {
        NaturalRunnable t1= new NaturalRunnable();
        Thread t2 = new Thread(t1);
        EvenRunnable t3= new EvenRunnable();
        Thread t4 = new Thread(t3);
        OddRunnable t5= new OddRunnable();
        Thread t6 = new Thread(t5);

        t2.start();
        t4.start();
        t6.start();
    }
}
```

## Lab 7. Write a Java program to demonstrate access modifiers.

### A. //save as A.java

```
package xxx;

public class A
{
    private int a = 10;
    int b = 20;           // Default
    protected int c = 30;
    public int d = 40;

    void printdata()
    {
        System.out.println("Private = " + a);
        System.out.println("Default = " + b);
        System.out.println("Protected = " + c);
        System.out.println("Public = " + d);
    }
}

class B extends A
{
    void display()
    {
        //System.out.println("Private = " + a);    private variable not accessible in same package
    }
}

class packdemo1
{
    public static void main(String args[])
    {
        B x=new B();
        x.printdata();
        x.display();
    }
}
```

### B.

```
package xxx;

public class packdemo2
{
    public static void main(String args[])
    {
        A x=new A();
        //System.out.println("Private = " + x.a);    private variable not accessible in same package
    }
}
```

```

non subclass
    System.out.println("Default = " + x.b);
    System.out.println("Protected = " + x.c);
    System.out.println("Public = " + x.d);
}
}

```

### C.

```

package yyy;
import xxx.A;

class AA extends A
{
    void print()
    {
        //System.out.println("Private = " + a);    private variable not accessible in different
package subclass
        //System.out.println("Default = " + b);    default variable not accessible in different
package subclass
        System.out.println("Protected = " + c);
        System.out.println("Public = " + d);
    }
}

public class packdemo3
{
    public static void main(String[] args)
    {
        AA x=new AA();
        x.print();
    }
}

```

### D.

```

package yyy;
import xxx.A;

public class packdemo4
{
    public static void main(String args[])
    {
        A x=new A();
        //System.out.println("Private = " + x.a);    private variable not accessible in different
package non subclass
        //System.out.println("Default = " + x.b);    default variable not accessible in
different package non subclass
        //System.out.println("Protected = " + x.c);    protected variable not accessible in
different package non subclass
        System.out.println("Public = " + x.d);
    }
}

```

## Lab 8. Write a Java program to demonstrate Swings (Arithmetic Operations).

```
package programs;

import java.awt.EventQueue;
import javax.swing.JOptionPane;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JTextField;
import javax.swing.JButton;
import java.awt.event.ActionListener;
import java.awt.event.ActionEvent;

public class swing1 extends JFrame {

    private static final long serialVersionUID = 1L;
    private JTextField tnum1;
    private JTextField tnum2;
    private JTextField tres;

    public static void main(String[] args) {
        EventQueue.invokeLater(new Runnable() {
            public void run() {
                try {
                    swing1 frame = new swing1();
                    frame.setVisible(true);
                } catch (Exception e) {
                    e.printStackTrace();
                }
            }
        });
    }

    public swing1() {

        setTitle("Simple Calculator");
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setBounds(100, 100, 520, 350);
        getContentPane().setLayout(null);

        // Label Num1
        JLabel lblNum1 = new JLabel("Num1");
        lblNum1.setBounds(70, 50, 60, 20);
        getContentPane().add(lblNum1);

        tnum1 = new JTextField();
        tnum1.setBounds(150, 50, 120, 25);
        getContentPane().add(tnum1);

        // Label Num2
        JLabel lblNum2 = new JLabel("Num2");
        lblNum2.setBounds(70, 90, 60, 20);
        getContentPane().add(lblNum2);

        tnum2 = new JTextField();
```

```

tnum2.setBounds(150, 90, 120, 25);
getContentPane().add(tnum2);

// Label Result
JLabel lblResult = new JLabel("Result");
lblResult.setBounds(70, 130, 60, 20);
getContentPane().add(lblResult);

tres = new JTextField();
tres.setBounds(150, 130, 120, 25);
tres.setEditable(false);
getContentPane().add(tres);

// ADD Button
JButton btnAdd = new JButton("Add");
btnAdd.setBounds(40, 190, 90, 30);
btnAdd.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        int n1 = Integer.parseInt(tnum1.getText());
        int n2 = Integer.parseInt(tnum2.getText());
        int res = n1 + n2;
        tres.setText(String.valueOf(res));
    }
});
getContentPane().add(btnAdd);

// SUB Button
JButton btnSub = new JButton("Sub");
btnSub.setBounds(150, 190, 90, 30);
btnSub.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        int n1 = Integer.parseInt(tnum1.getText());
        int n2 = Integer.parseInt(tnum2.getText());
        int res = n1 - n2;
        tres.setText(String.valueOf(res));
    }
});
getContentPane().add(btnSub);

// MUL Button
JButton btnMul = new JButton("Mul");
btnMul.setBounds(260, 190, 90, 30);
btnMul.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        int n1 = Integer.parseInt(tnum1.getText());
        int n2 = Integer.parseInt(tnum2.getText());
        int res = n1 * n2;
        tres.setText(String.valueOf(res));
    }
});
getContentPane().add(btnMul);

// DIV Button
JButton btnDiv = new JButton("Div");
btnDiv.setBounds(370, 190, 90, 30);
btnDiv.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        int n1 = Integer.parseInt(tnum1.getText());

```

```

        int n2 = Integer.parseInt(tnum2.getText());

        if (n2 == 0) {
            JOptionPane.showMessageDialog(null, "Cannot divide by zero!");
        } else {
            double res = (double) n1 / n2;
            tres.setText(String.valueOf(res));
        }
    }

});
getContentPane().add(btnDiv);

// MOD Button
JButton btnMod = new JButton("Mod");
btnMod.setBounds(150, 240, 90, 30);
btnMod.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        int n1 = Integer.parseInt(tnum1.getText());
        int n2 = Integer.parseInt(tnum2.getText());

        if (n2 == 0) {
            JOptionPane.showMessageDialog(null, "Cannot perform modulus by zero!");
        } else {
            int res = n1 % n2;
            tres.setText(String.valueOf(res));
        }
    }
});
getContentPane().add(btnMod);

// CLEAR Button
JButton btnClear = new JButton("Clear");
btnClear.setBounds(260, 240, 90, 30);
btnClear.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        tnum1.setText("");
        tnum2.setText("");
        tres.setText("");
        tnum1.requestFocus();
    }
});
getContentPane().add(btnClear);
}
}

```